

LAEGEREN, Switzerland

WMO: 066690

Lat: 47.4819N

Lon: 8.3972E

Elev: 843

StdP: 91.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.5	(c) -8.4	(d) -12.8	(e) 1.4	(f) -7.5	(g) -10.7	(h) 1.7	(i) -6.0	(j) 13.6	(k) 3.7	(l) 12.4	(m) 3.8	(n) 3.1	(o) 30	(p) 0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.8	(c) 26.9	(d) 17.8	(e) 25.1	(f) 17.0	(g) 23.5	(h) 16.5	(i) 18.7	(j) 24.6	(k) 17.9	(l) 23.4	(m) 17.2	(n) 22.2	(o) 1.4	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.6	(b) 13.1	(c) 21.1	(d) 15.9	(e) 12.5	(f) 20.2	(g) 15.2	(h) 11.9	(i) 19.3	(j) 56.8	(k) 24.8	(l) 54.0	(m) 23.4	(n) 51.6	(o) 22.3	(p) 23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.9	9.4	8.2	-12.0	29.1	2.8	2.0	-14.0	30.5	-15.6	31.7	-17.1	32.8	-19.1	34.2
			-12.3	20.3	2.7	1.0	-14.2	21.0	-15.8	21.6	-17.3	22.2	-19.3	23.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-0.8	0.1	3.8	7.6	11.6	15.3	17.1	16.8	12.6	8.7	3.5	0.3
	DBStd	7.61	4.16	4.70	4.48	4.62	4.13	4.27	3.78	3.82	3.68	4.10	4.20	4.09
	HDD10.0	1535	336	278	198	100	30	5	1	1	16	73	197	302
	HDD18.3	3823	594	512	452	323	211	110	70	77	174	297	444	560
	CDD10.0	837	0	0	4	27	81	164	219	212	93	34	3	0
	CDD18.3	84	0	0	0	0	3	19	31	29	2	0	0	0
	CDH23.3	382	0	0	0	1	7	87	148	138	2	0	0	0
	CDH26.7	58	0	0	0	0	0	11	23	23	0	0	0	0
(14) Wind	WSAvg	2.9	3.4	3.4	3.3	3.1	2.7	2.5	2.5	2.4	2.7	2.9	3.2	3.3
(15) Precipitation	PrecAvg	1131	70	69	71	84	121	124	127	119	86	86	88	87
	PrecMax	1455	170	166	233	168	231	207	227	221	174	153	193	185
	PrecMin	845	13	20	24	9	31	50	43	19	39	10	4	0
	PrecStd	155	43	39	46	42	52	42	48	46	36	39	47	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	11.2	15.9	20.9	24.3	28.3	29.1	29.2	23.1	19.0	14.4	10.6
		MCWB	6.0	5.7	9.0	11.5	15.1	18.4	18.3	17.5	16.3	12.9	8.9	6.1
	2%	DB	7.5	9.3	13.7	18.5	22.1	25.8	26.8	26.7	21.5	17.1	12.5	8.2
		MCWB	4.9	4.9	7.5	10.4	14.2	17.2	17.7	17.8	15.6	12.6	8.4	5.7
	5%	DB	6.3	7.7	11.9	16.5	20.1	23.9	24.9	24.6	19.8	15.7	10.7	6.9
		MCWB	4.1	4.2	6.8	9.7	13.1	16.6	17.0	17.1	15.0	12.1	7.9	4.9
	10%	DB	4.8	6.2	10.1	14.6	18.2	22.0	23.1	22.9	18.2	14.3	9.2	5.7
		MCWB	3.1	3.6	6.0	8.8	12.2	15.8	16.4	16.7	14.3	11.6	7.2	4.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.0	9.6	12.4	16.4	19.8	19.8	19.6	17.5	14.6	10.6	7.8
		MCDB	8.1	8.7	13.9	18.8	21.9	26.6	26.6	25.0	21.4	17.0	12.2	8.6
	2%	WB	5.9	5.6	8.5	11.1	15.1	18.3	18.6	18.6	16.5	13.5	9.3	6.5
		MCDB	6.7	8.0	12.1	16.8	20.3	23.8	24.6	24.4	20.0	15.8	11.1	7.6
	5%	WB	4.5	4.8	7.6	10.2	13.9	17.2	17.7	17.8	15.5	12.8	8.4	5.5
		MCDB	5.7	7.0	10.7	15.3	18.7	22.4	23.2	23.2	18.8	15.0	10.1	6.7
	10%	WB	3.3	3.9	6.5	9.2	12.8	16.3	17.0	17.1	14.6	11.9	7.2	4.4
		MCDB	4.5	5.8	9.5	13.8	17.3	21.0	22.1	22.0	17.6	13.8	9.0	5.5
(35) Mean Daily Temperature Range	MDBR		3.6	4.1	5.5	6.6	6.8	6.8	6.8	6.3	5.5	4.6	3.8	3.7
	5% DB	MCDBR	4.9	6.2	7.6	8.8	8.7	8.7	8.9	8.4	7.3	5.9	5.9	5.0
		MCWBR	3.4	3.6	3.9	4.3	4.4	4.2	3.9	3.7	3.7	3.3	3.1	3.3
	5% WB	MCDBR	4.4	5.4	6.9	8.2	8.0	8.1	8.2	7.8	6.6	5.4	5.2	4.3
		MCWBR	3.7	3.7	4.0	4.2	4.4	4.2	4.0	3.8	3.6	3.4	3.4	3.5
(40) Clear-Sky Solar Irradiance	taub		0.281	0.304	0.336	0.364	0.376	0.380	0.381	0.374	0.352	0.332	0.302	0.277
	taud		2.480	2.420	2.383	2.331	2.332	2.355	2.341	2.375	2.423	2.470	2.507	2.482
	Ebn at Noon		818	865	885	888	888	883	876	868	856	820	782	776
	Edh at Noon		63	83	101	117	121	120	120	111	96	79	62	56
(44) All-Sky Solar Radiation	RadAvg		1.14	1.97	3.18	4.33	5.00	5.75	5.55	4.84	3.63	2.27	1.25	0.93
	RadStd		0.13	0.27	0.43	0.62	0.50	0.54	0.51	0.50	0.37	0.24	0.15	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.63	+0.43	N/A	N/A	+75	+29
(47) Regional (116 neighbors)	N/A	N/A	N/A	+0.70	+0.42	+0.34	N/A	N/A	+56	+15

Nomenclature: See separate page